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CT Reconstruction with Learned Priors for Low-Dose Imaging

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July 2026

Submitted in partial fulfillment of the requirements for the Master of Philosophy in Data Intensive Science

Total page count: ??

Main chapters (excluding front-matter, references and appendix): 1 pages (pp ??–??)

Main chapters word count: 0

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Abstract

This project investigates learned priors for low-dose computed tomography reconstruction, with particular focus on stability, regularisation behaviour, and quantitative performance relative to classical reconstruction approaches.

Acknowledgements

Optional acknowledgements go here.

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Introduction

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Background

This chapter establishes the mathematical and physical foundations required to understand learned reconstruction methods for computed tomography (CT). We begin with the X-ray physics and the forward model, move to a careful treatment of ill-posedness, and conclude with a survey of the classical reconstruction algorithms that serve as baselines throughout this project.

2.1 Computed Tomography: Physical and Mathematical Foundations

2.1.1 X-ray Attenuation and the Beer–Lambert Law

When a monochromatic X-ray beam of initial intensity I_0 traverses a material along a straight line L, photons are absorbed and scattered by the medium. The local probability of an interaction per unit length is described by the *linear attenuation coefficient* $\mu(\boldsymbol{x}) \geq 0$, which depends on both tissue composition and photon energy. The Beer–Lambert law governs the transmitted intensity I:

$$I = I_0 \exp\left(-\int_L \mu(\boldsymbol{x})\,\mathrm{d}\ell\right), \tag{2.1}$$

where ℓ parameterises arc length along the ray. Taking logarithms, one obtains the *line integral* (or projection):

$$p = -\ln\left(\frac{I}{I_0}\right) = \int_L \mu(\boldsymbol{x})\,\mathrm{d}\ell. \tag{2.2}$$

In practice, μ is the unknown quantity one wishes to recover; p is measurable for each ray. A full CT acquisition collects such integrals over many lines, indexed by angle ϕ and detector position s, forming the *sinogram*.

2.1.2 The Continuous Radon Transform

The mathematical abstraction of the projection process is the *Radon transform* (?). For a function $f : \mathbb{R}^2 \to \mathbb{R}$ (representing the attenuation map), the Radon transform $\mathcal{R}$ is defined as:

$$(\mathcal{R}f)(\phi, s) = \int_{-\infty}^{\infty} f\big(s\boldsymbol{\theta} + t\boldsymbol{\theta}^{\perp}\big)\,\mathrm{d}t, \tag{2.3}$$

where $\boldsymbol{\theta} = (\cos\phi, \sin\phi)^T$ is the unit vector in the projection direction, $\boldsymbol{\theta}^{\perp} = (-\sin\phi, \cos\phi)^T$ is orthogonal to it, and $s \in \mathbb{R}$ is the signed distance from the origin to the line. The domain of (ϕ, s) is called the *sinogram domain* or *Radon domain.* For a three-dimensional object, the corresponding transform integrates over planes rather than lines; in fan-beam or cone-beam geometries the lines emanate from a single source point, yielding a modified geometry (?).

A key analytical result is the *Central Slice Theorem* (also called the Fourier Slice Theorem): the one-dimensional Fourier transform of a projection at angle ϕ equals a radial slice of the two-dimensional Fourier transform of f at the same angle. This relationship underpins the filtered backprojection algorithm described in **??**.

2.1.3 The Discrete Forward Model

Physical CT systems are digital: the detector has a finite number of pixels, the source rotates to a finite set of angles, and the reconstructed image is represented on a grid. After discretisation, the continuous relation (**??**) becomes a large linear system. Let $\boldsymbol{x} \in \mathbb{R}^n$ be the vectorised image (pixel values of the attenuation map arranged in raster order) and $\boldsymbol{y} \in \mathbb{R}^m$ be the vector of all measured line integrals (the sinogram, row-stacked). The *forward model* is:

$$\boldsymbol{y} = A\boldsymbol{x} + \boldsymbol{\eta}, \tag{2.4}$$

where $A \in \mathbb{R}^{m\times n}$ is the *system matrix* (or *projection matrix*) whose entry A_{ij} encodes the contribution of pixel j to measurement i—typically the length of the intersection of ray i with pixel j—and $\boldsymbol{\eta}$ represents additive noise. This is the fundamental model throughout the project; the goal of reconstruction is to recover $\boldsymbol{x}$ from $\boldsymbol{y}$ and A.

Remark 2.1. *In the LION/tomosipo framework used in this project (?), A is never stored explicitly—it would be $\mathcal{O}(mn)$ in memory for large volumes—but is instead evaluated on-the-fly via GPU-accelerated forward and back-projection operators. The linear map A and its adjoint A^T (the* backprojection *operator) are therefore available as callable functions rather than dense matrices.*

2.2 Ill-Posedness of the CT Inverse Problem

Recovering $\boldsymbol{x}$ from $\boldsymbol{y}$ is an *inverse problem.* When measurements are complete and noise-free, the Radon transform is invertible (via the classical inversion formula); however, figures/sinogram_comparison.pdf

Figure 2.1: Sinogram comparison for the Shepp-Logan phantom. **Left**: full sinogram with 720 projection angles (the clinical standard). **Right**: sparse sinogram with only 30 angles, corresponding to a $24\times$ dosage reduction. The sparse sinogram contains the same sinusoidal structure but sampled at far fewer angular positions, leaving large gaps in the Radon domain.

practical CT acquisition violates both conditions, rendering the problem ill-posed in the sense of Hadamard (**?**): solutions may not exist, may not be unique, or may not depend continuously on the data.

2.2.1 Sources of Ill-Posedness

Sparse-view CT. To reduce patient radiation dose—the primary concern motivating this project—one reduces the number of projection angles. The Mayo Clinic Low Dose CT Grand Challenge dataset (**?**) provides a canonical low-dose benchmark. When the number of angles drops below the Nyquist-like sampling criterion (informally: at least $(\pi/2)N$ views for an $N \times N$ image), the sinogram is severely undersampled. Analytically, the null space of the undersampled A is non-trivial: many images $\boldsymbol{x}$ yield the same measurements $\boldsymbol{y}$, so the problem is underdetermined ($m \ll n$). The artefacts produced by naive reconstruction are radially symmetric streak artefacts aligned with the missing-angle directions.

Limited-angle CT. A related but distinct scenario arises when projections are available only over a restricted angular range (e.g., $[0°, 120°]$). In tomosynthesis and some industrial settings this is unavoidable due to geometry constraints. Here the null space of A contains low-frequency components aligned with the missing angular sector, producing characteristic directional blurring.

Noise amplification. Even with a full set of angles, the continuous Radon inversion formula amplifies high-frequency noise: it involves differentiation (the ramp filter in FBP), which multiplies Fourier coefficients by $|\xi|$, the spatial frequency. Consequently, white measurement noise $\boldsymbol{\eta}$ leads to high-frequency artefacts in naively inverted images. Physically, quantum (Poisson) noise in the photon counts is the dominant source (**?**).

Formal characterisation. The singular value decomposition (SVD) of A reveals the mechanism: the singular values σ_i of A decay, often rapidly, towards zero. Inverting A via $\boldsymbol{x} = A^+\boldsymbol{y}$ (the pseudoinverse) amplifies the components of $\boldsymbol{y}$ lying in the direction of small singular vectors by $1/\sigma_i$, which diverges as $\sigma_i \to 0$. This is the hallmark of a *discrete ill-posed problem* (**??**).

2.2.2 Noise Model

The physical acquisition process introduces Poisson-distributed noise in the photon counts. After the log-transformation (**??**), the sinogram noise is approximately Gaussian for moderate dose levels (by the central limit theorem) and can be written:

$$\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \sigma^2 I), \tag{2.5}$$

figures/reconstruction_comparison.pdf

Figure 2.2: Reconstruction comparison on the Shepp-Logan phantom with 5% Gaussian noise. **Top row**: ground truth, FBP from 720 views (clinical reference), and FBP from 30 views (severe streak artefacts). **Bottom row**: SIRT, NAG_LS, and TV minimisation, all from 30 views. The degradation from 720 to 30 views in FBP illustrates the ill-posedness of the sparse-view problem.

where σ^2 depends on dose level. In the sparse-view CT experiments of the TFPnP paper (?), Gaussian noise with levels $\sigma \in \{5\%, 7.5\%, 10\%\}$ is added to the projections after the Radon transform, simulating realistic low-dose acquisition at a 24× dosage reduction compared to the 720-view standard.

2.3 Classical Reconstruction Algorithms

Before learned methods, three families of algorithms dominated CT reconstruction. These serve as baselines in the experimental evaluation and underpin the learned methods reviewed in ??.

2.3.1 Filtered Backprojection (FBP)

Filtered backprojection is the standard clinical workhorse, derived analytically from the inversion of the continuous Radon transform via the Central Slice Theorem. The algorithm proceeds in two stages:

1. **Filtering.** Each projection $p(\phi, \cdot)$ is convolved in the s-domain with a *ramp filter* $h(s)$ whose Fourier transform is $|\xi|$, compensating for the non-uniform density of Fourier samples:
$$\tilde{p}(\phi, s) = \big(p(\phi, \cdot) * h\big)(s).$$
2. **Backprojection.** The filtered projections are smeared back into image space by integrating over all angles:
$$f(\boldsymbol{x}) = \int_0^{\pi} \tilde{p}(\phi,\, \boldsymbol{x} \cdot \boldsymbol{\theta})\, \mathrm{d}\phi.$$

FBP is exact in the limit of dense, noiseless, full-angle sampling. Its principal shortcomings are: (i) the ramp filter amplifies high-frequency measurement noise, requiring an apodisation window (e.g., Hann, Ram-Lak) that blurs fine structures; (ii) streak artefacts appear under sparse-view sampling. For cone-beam geometry, the Feldkamp–Davis–Kress (FDK) algorithm (?) extends FBP and is available in LION as `fdk`.

2.3.2 Simultaneous Iterative Reconstruction Technique (SIRT)

Iterative algebraic methods treat reconstruction as a least-squares problem:

$$\min_{\boldsymbol{x}} \frac{1}{2} \| A\boldsymbol{x} - \boldsymbol{y} \|_2^2. \tag{2.6}$$

SIRT (?) approximates the solution via gradient descent with a scaled step:

$$\boldsymbol{x}^{(k+1)} = \boldsymbol{x}^{(k)} + CA^T R\big(\boldsymbol{y} - A\boldsymbol{x}^{(k)}\big)\,, \tag{2.7}$$

where $C = \mathrm{diag}(1/\sum_i A_{ij})$ and $R = \mathrm{diag}(1/\sum_j A_{ij})$ are diagonal scaling matrices (column and row sums of A) that normalise the step and ensure convergence. SIRT converges to the minimum ℓ^2-norm least-squares solution. It handles missing data more gracefully than FBP (no explicit ramp amplification), but converges slowly and is still unstable in the presence of noise without early stopping or regularisation. LION provides a differentiable implementation via `ts_algorithms.SIRT`.

2.3.3 Total Variation Minimisation

Motivated by the observation that medical CT images are approximately piecewise constant (and thus have sparse gradients), **?** introduced the *total variation* (TV) regulariser:

$$\mathrm{TV}(\boldsymbol{x}) = \|\nabla \boldsymbol{x}\|_1 = \sum_{i,j} \sqrt{|\partial_x x_{ij}|^2 + |\partial_y x_{ij}|^2}, \tag{2.8}$$

and **?** applied it to CT reconstruction as:

$$\min_{\boldsymbol{x} \geq 0} \mathrm{TV}(\boldsymbol{x}) \quad \text{subject to} \quad \|A\boldsymbol{x} - \boldsymbol{y}\|_2 \leq \varepsilon, \tag{2.9}$$

where ε bounds the noise level. This is a convex optimisation problem solvable by primal-dual splitting methods, such as the Primal-Dual Hybrid Gradient (PDHG) algorithm of **?**, available in LION as `tv_min`. TV reconstruction produces visually clean images with preserved edges, but can introduce the characteristic *staircase artefact* and requires careful selection of λ (or equivalently ε). In the TFPnP sparse-view CT experiments, TV with optimally tuned parameters achieves 20.91/19.17/16.95 dB PSNR for the three noise levels—substantially above FBP but below learned methods (**?**).

2.3.4 From Classical to Learned: A Bridge

The fundamental limitation of both FBP and iterative methods is that they either (a) invert the forward model analytically without data-driven priors (FBP), or (b) assume a hand-crafted regulariser such as TV whose inductive bias may not align with the true distribution of CT images. The key question motivating the field of learned reconstruction is: *can we replace or supplement the hand-crafted prior with one learned directly from data?* **??** surveys the spectrum of approaches that have emerged to address this question—from variational regularisation through deep unrolling to plug-and-play methods—culminating in TFPnP, the method implemented in this project.

figures/difference_maps.pdf

Figure 2.3: Difference maps $\hat{\boldsymbol{x}} - \boldsymbol{x}^*$ for three reconstruction methods from 30-view sparse sinograms with 5% noise. **Left**: FBP shows structured streak artefacts (correlated error). **Centre**: SIRT shows diffuse error concentrated at edges. **Right**: TV minimisation achieves the lowest error overall but shows sharp transitions at edges characteristic of the staircase artefact.

Chapter 3

Related Work

This chapter positions the paper at the centre of this project—*TFPnP: Tuning-free Plug-and-Play Proximal Algorithms with Applications to Inverse Imaging Problems* (**?**)—within the broader landscape of CT reconstruction methods. We trace the intellectual lineage from classical iterative methods through variational regularisation and plug-and-play frameworks to the deep learning methods against which TFPnP is benchmarked. We then provide a detailed technical and critical analysis of the paper itself.

3.1 Classical Iterative Methods

As described in **??**, iterative methods such as SIRT, ART (Algebraic Reconstruction Technique) (**?**), and CGLS (Conjugate Gradient Least Squares) minimise the data-fidelity term $\frac{1}{2}\|A\boldsymbol{x} - \boldsymbol{y}\|_2^2$ without an explicit regulariser. Their convergence properties are well understood, but they are limited by the ill-posedness of the inverse problem: without regularisation, they overfit to noise given sufficiently many iterations. Early stopping acts as an implicit regulariser, introducing a trade-off between data consistency and noise suppression that is difficult to tune automatically.

3.2 Variational Regularisation

Variational methods frame reconstruction as a regularised optimisation problem:

$$\hat{\boldsymbol{x}} = \arg\min_{\boldsymbol{x}} \; D(\boldsymbol{y}, A\boldsymbol{x}) + \lambda R(\boldsymbol{x}), \tag{3.1}$$

where D is a data-fidelity term (e.g., $\frac{1}{2}\|\cdot\|_2^2$ for Gaussian noise, or a Kullback–Leibler divergence for Poisson noise) and R is a regulariser that encodes prior knowledge about $\boldsymbol{x}$. The parameter $\lambda > 0$ controls the trade-off between data fit and regularity. A rich hierarchy of regularisers has been studied (**?**):

- **Tikhonov regularisation** (?): $R(\boldsymbol{x}) = \frac{1}{2}\|\boldsymbol{x}\|_2^2$. Penalises energy; yields smooth solutions; closed-form inversion via $(A^TA + \lambda I)^{-1}A^T\boldsymbol{y}$.
- **Total variation** (?): $R(\boldsymbol{x}) = \|\nabla\boldsymbol{x}\|_1$. Promotes piecewise-constant images; well-suited to CT; solved by PDHG or ADMM.
- **Sparsity-promoting regularisers**: Wavelet ℓ^1 penalties (?), dictionary learning (?), and low-rank regularisation (?).

The key limitation of all explicit regularisers is that their functional forms embody strong modelling assumptions—piecewise constancy for TV, wavelet sparsity for wavelet penalties—that may not hold for complex anatomical structures. Furthermore, λ must be chosen per-problem, a process that is non-trivial in clinical practice.

3.3 Proximal Algorithms

When R is non-smooth (as for TV), standard gradient descent cannot be applied. *Proximal algorithms* decompose the composite minimisation into subproblems involving the *proximal operator* (??):

$$\mathrm{Prox}_{\gamma R}(\boldsymbol{v}) = \arg\min_{\boldsymbol{x}} \ \gamma R(\boldsymbol{x}) + \frac{1}{2}\|\boldsymbol{x} - \boldsymbol{v}\|_2^2. \tag{3.2}$$

For many regularisers, (??) has a closed-form solution (e.g., soft-thresholding for $R = \|\cdot\|_1$).

Proximal gradient descent (Forward–Backward splitting). When D is smooth with L-Lipschitz gradient and R is non-smooth, the iterates are:

$$\boldsymbol{x}^{k+1} = \mathrm{Prox}_{\gamma R}\big(\boldsymbol{x}^k - \gamma\nabla D(\boldsymbol{x}^k)\big)\,. \tag{3.3}$$

FISTA (?) adds Nesterov momentum to achieve $\mathcal{O}(1/k^2)$ convergence.

ADMM. When both D and R are non-smooth, the Alternating Direction Method of Multipliers (ADMM) (??) introduces an auxiliary variable $\boldsymbol{z}$ to decouple D and R:

$$\boldsymbol{x}^{k+1} = \mathrm{Prox}_{\sigma^2 R}(\boldsymbol{z}^k - \boldsymbol{u}^k), \tag{3.4}$$

$$\boldsymbol{z}^{k+1} = \mathrm{Prox}_{\frac{1}{\mu}D}(\boldsymbol{x}^{k+1} + \boldsymbol{u}^k), \tag{3.5}$$

$$\boldsymbol{u}^{k+1} = \boldsymbol{u}^k + \boldsymbol{x}^{k+1} - \boldsymbol{z}^{k+1}, \tag{3.6}$$

where $\mu > 0$ is a penalty parameter and $\sigma^2 = \lambda/\mu$. ADMM alternates between an image-domain regularisation update (??), a data-consistency update (??), and a dual variable update (??).

3.4 Plug-and-Play (PnP) Methods

The *Plug-and-Play* (PnP) framework (?) is the immediate precursor to TFPnP and deserves careful treatment.

3.4.1 The PnP Insight

The $\boldsymbol{x}$-update in ADMM (??) is precisely the proximal operator of R applied to $\boldsymbol{z}^k - \boldsymbol{u}^k$. By (??), this is equivalent to a denoising problem: given a noisy observation $\boldsymbol{v} = \boldsymbol{z}^k - \boldsymbol{u}^k$, find the image that minimises $\sigma^2 R(\boldsymbol{x}) + \frac{1}{2}\|\boldsymbol{x} - \boldsymbol{v}\|_2^2$. The PnP insight is to replace this proximal step with *any* off-the-shelf denoiser $\mathcal{H}_\sigma$, regardless of whether it corresponds to an explicit R:

$$\boldsymbol{x}^{k+1} = \mathcal{H}_{\sigma_k}(\boldsymbol{z}^k - \boldsymbol{u}^k). \tag{3.7}$$

The resulting *PnP-ADMM* scheme is:

$$\boldsymbol{x}^{k+1} = \mathcal{H}_{\sigma_k}(\boldsymbol{z}^k - \boldsymbol{u}^k), \tag{3.8}$$

$$\boldsymbol{z}^{k+1} = \mathrm{Prox}_{\frac{1}{\mu_k}D}(\boldsymbol{x}^{k+1} + \boldsymbol{u}^k), \tag{3.9}$$

$$\boldsymbol{u}^{k+1} = \boldsymbol{u}^k + \boldsymbol{x}^{k+1} - \boldsymbol{z}^{k+1}. \tag{3.10}$$

This framework decouples the forward model (handled in the $\boldsymbol{z}$-step) from the image prior (handled via the denoiser in the $\boldsymbol{x}$-step), allowing state-of-the-art denoisers—BM3D (?), NLM, or deep CNN denoisers such as DnCNN (?)—to be swapped in without retraining for each new inverse problem.

3.4.2 PnP Variants and Extensions

The body of PnP literature extends the framework in several directions (?):

- **Proximal algorithms.** PnP has been combined with ADMM (?), half-quadratic splitting (HQS) (?), proximal gradient (PnP-PGM) (?), accelerated proximal gradient (PnP-APGM) (?), and their stochastic variants.

- **Regularisation by Denoising (RED).** ? proposed an explicit regulariser $R(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^T(\boldsymbol{x} - \mathcal{H}_\sigma(\boldsymbol{x}))$ whose gradient $\nabla R(\boldsymbol{x}) = \boldsymbol{x} - \mathcal{H}_\sigma(\boldsymbol{x})$ is the denoising residual, enabling gradient-based optimisation.

- **Denoising-based AMP (D-AMP).** ? adapts the Approximate Message Passing framework by replacing its thresholding step with a deep denoiser, exploiting the Onsager correction to track effective noise statistics.

- **Theoretical convergence.** Convergence of PnP algorithms is challenging to prove because denoisers do not generally correspond to the proximal operator of any convex function. Progress has been made under assumptions of bounded or firmly non-expansive denoisers (?).

3.4.3 The Parameter Tuning Problem

Despite their empirical success, PnP algorithms suffer a fundamental practical limitation: their performance is highly sensitive to the choice of internal parameters—principally the denoising strength σ_k at each iteration, the penalty parameter μ_k, and the termination time τ. As demonstrated in Figure 1 of **?**, the optimal PSNR for each image in a dataset occurs at a *different* parameter setting, and sometimes even requires early stopping to avoid quality degradation. No single fixed parameter schedule achieves uniformly good performance across images with varying noise levels and content. This *manual tuning problem* is the precise gap that TFPnP is designed to fill.

3.5 Algorithm Unrolling and Learned Primal Dual

Algorithm unrolling (?) is a complementary paradigm that unrolls a fixed number of optimisation iterations into a deep neural network, replacing hand-crafted components with learned ones and training the entire unrolled system end-to-end. Learned Primal Dual (LPD) (?) is the leading algorithm-unrolling method for CT reconstruction and the primary benchmark against which TFPnP is compared.

LPD replaces the proximal operators of a primal-dual algorithm with small CNNs:

$$h^{k+1} = \Gamma_\phi^k\big(h^k,\ A\boldsymbol{x}^k,\ \boldsymbol{y}\big)\,, \tag{3.11}$$

$$g^{k+1} = \Lambda_\theta^k\big(g^k,\ A^T h^{k+1}\big)\,, \tag{3.12}$$

$$\boldsymbol{x}^{k+1} = \boldsymbol{x}^k + g^{k+1}, \tag{3.13}$$

where Γ_ϕ^k and Λ_θ^k are small CNNs (with unshared weights across iterations) operating in measurement space and image space respectively, and h^k, g^k are running memory variables. The network is trained end-to-end by minimising $\|\boldsymbol{x}^K - \boldsymbol{x}^*\|_2^2$ over paired training data $(\boldsymbol{y}, \boldsymbol{x}^*)$.

LPD is state-of-the-art in supervised CT reconstruction performance, achieving 22.72/22.95/22.50 dB PSNR at the three noise levels in the TFPnP benchmark (?). However, it requires a separate trained model for each acquisition geometry and noise level; it cannot generalise across measurement conditions without retraining, unlike PnP methods.

3.6 Deep Learning for CT Reconstruction

The broader field of deep learning for CT reconstruction can be taxonomised along two axes: (i) the degree to which the forward model A is incorporated into the architecture, and (ii) whether the method is supervised or unsupervised. The benchmarks used in the TFPnP CT experiments (?) span this taxonomy:

- **FBP + post-processing (FBPconv)** (?): applies FBP to produce an initial reconstruction, then trains a CNN to remove artefacts in image space. Fast but does not enforce data consistency. Achieves 22.16 dB.
- **RED-CNN** (?): a CNN trained in image space using residual learning to denoise low-dose CT reconstructions. Achieves 22.34 dB. Does not use the forward model during inference.
- **RPGD** (?): combines projected gradient descent steps (using A and A^T) with a CNN denoiser trained as a proximal operator. Enforces data consistency; achieves 23.19 dB.
- **LPD** (?): as described in **??**; achieves 22.72 dB.

3.7 TFPnP: Tuning-free Plug-and-Play Proximal Algorithm

3.7.1 Paper Overview and Positioning

The paper "TFPnP: Tuning-free Plug-and-Play Proximal Algorithms with Applications to Inverse Imaging Problems" (?) was published in the *Journal of Machine Learning Research* (Volume 23, 2022) by Kaixuan Wei, Angelica Aviles-Rivero, Jingwei Liang, Ying Fu, Hua Huang, and Carola-Bibiane Schönlieb. It extends a conference paper at ICML 2020 (?).

The paper sits at the intersection of three traditions: PnP methods (exploiting the modular separation of forward model and denoiser prior), reinforcement learning (used as the optimisation engine for parameter search), and inverse imaging (validated on CS-MRI, sparse-view CT, single-photon imaging, and phase retrieval). Its core thesis is that the internal parameter selection in PnP algorithms—previously treated as manual hyperparameters or governed by handcrafted heuristics—can be framed as a *sequential decision-making problem* and solved by a learned policy network.

3.7.2 Main Contributions

The paper makes three principal contributions (?):

1. **The TFPnP framework.** A class of PnP algorithms in which the parameter selection (denoising strength σ_k, penalty μ_k, and termination time τ) is governed by a learned policy π, rather than fixed or handcrafted schedules. The framework

is generic: it is instantiated for PnP-ADMM, PnP-HQS, PnP-PGM, PnP-APGM, RED-ADMM, and D-AMP.

2. **A mixed model-free and model-based RL algorithm.** The policy network has two sub-policies: a stochastic sub-policy π_1 for the discrete termination decision (stop or continue), trained with model-free policy gradients; and a deterministic sub-policy π_2 for the continuous parameters (σ_k, μ_k), trained via model-based back-propagation through the differentiable PnP-ADMM iterations.

3. **Extensive empirical validation.** State-of-the-art results on CS-MRI, sparse-view CT, single-photon imaging, and phase retrieval, with ablations of the denoiser prior choice, policy design, and hyperparameter sensitivity.

3.7.3 Technical Details: The RL Formulation

The TFPnP algorithm is built on a Markov Decision Process (MDP) formulation (S, A, p, r) (**?**):

- **State space $\mathcal{S}$.** The current optimisation state $s^k = (\boldsymbol{x}^k, \boldsymbol{z}^k, \boldsymbol{u}^k)$, augmented with scalars encoding the noise level and the iteration count.
- **Action space $\mathcal{A}$.** Actions are pairs (a_1, a_2) where $a_1 \in \{0, 1\}$ is the termination decision and $a_2 = (\sigma_k, \mu_k)$ are the continuous parameters.
- **Transition function p.** Defined by m consecutive PnP-ADMM iterations; differentiable with respect to the U-Net denoiser (**?**).
- **Reward function r.** A scalar measuring image quality improvement relative to the previous state, encouraging policies that reach high-quality reconstructions efficiently and terminate at the optimal iteration.

The objective is:

$$J(\pi) = \mathbb{E}_{s^0 \sim \mathcal{S}_0, \mathcal{T} \sim \pi}[R_0], \tag{3.14}$$

where $R_0 = \sum_{t'=0}^{N-1} \rho^{t'} r(s^{t+t'}, a^{t+t'})$ is the discounted return and $\rho \in [0, 1]$ is a discount factor. The policy and value networks are trained jointly: the value network V_ϕ^π minimises the Bellman residual, while π_1 is trained via the REINFORCE policy gradient theorem and π_2 is trained via deterministic policy gradient through the differentiable environment.

3.7.4 Key Innovations

Joint optimisation of discrete and continuous parameters. Prior automated parameter selection methods (L-curve (**?**), generalised cross-validation, Stein's unbiased risk estimator) targeted a single fixed parameter for all iterations. TFPnP produces a *schedule* of parameters adapting to the current optimisation state.

Model-based gradient estimation for continuous parameters. Using the differentiable U-Net denoiser, TFPnP backpropagates through the PnP iterations to compute the exact gradient of the return with respect to (σ_k, μ_k). This model-based gradient has far lower variance than Monte Carlo estimates, substantially accelerating training of the continuous sub-policy.

Generalisation across imaging conditions. Because the policy encodes the noise level and iteration count as auxiliary inputs, a *single* trained policy generalises across different noise levels and undersampling factors—a property that supervised end-to-end methods such as LPD lack by design.

Early stopping as a first-class decision. TFPnP treats the termination time as a learnable decision, allowing the policy to stop as soon as further iterations would degrade quality due to noise overfitting. This is a critical robustness property.

3.7.5 Experimental Results: Sparse-View CT

The CT benchmark uses 30-view parallel-beam projections (a 24× dosage reduction relative to the 720-view standard (**??**)) with Gaussian noise at three levels. Results are reported in **??** (**?**).

Table 3.1: Sparse-view CT PSNR (dB) comparison across all methods. TFPnP achieves the best result at every noise level. Data from **?**, Table 5.

Method	**5%**	**7.5%**	**10%**
FBP (**?**)	18.37	16.16	14.32
TV (**?**)	21.47	19.17	16.95
FBPconv (**?**)	23.00	22.94	22.86
RED-CNN (**?**)	23.45	23.11	22.71
LPD (**?**)	23.86	23.44	23.00
RPGD (**?**)	24.29	23.82	23.38
TFPnP (Ours)	**24.50**	**24.23**	**23.72**

TFPnP achieves the best results at all noise levels, outperforming the second-best method (RPGD) by 0.21–0.34 dB consistently across noise levels, suggesting the learned policy adapts effectively to varying signal-to-noise conditions.

3.7.6 Comparison to Prior Work

vs. classical PnP. Classical PnP-ADMM with a fixed parameter schedule (e.g., the IRCNN handcrafted schedule of **?**) achieves sub-optimal PSNR because no single schedule serves all images. TFPnP demonstrates that the learned policy reaches results close to the *oracle policy*—one that has access to the ground truth to set parameters—without any ground truth at test time.

vs. supervised deep learning (LPD, RED-CNN, FBPconv). Supervised methods require *paired* training data and produce a fixed function from sinogram to image. TFPnP's denoiser can be trained on image-space patches (unpaired with sinograms) and the RL policy is trained separately; this modular training is both more data-efficient and more interpretable. TFPnP also inherits data consistency from the $\boldsymbol{z}$-step (**??**), whereas image-domain methods (RED-CNN, FBPconv) do not enforce $\|A\boldsymbol{x} - \boldsymbol{y}\|_2 \leq \varepsilon$.

vs. algorithm unrolling (LPD). Both methods incorporate the forward operator A during inference. The key difference is that LPD's number of unrolled iterations is fixed at training time; TFPnP's termination time is adaptive and learned. Additionally, TFPnP's policy can adapt to unseen noise levels, whereas LPD requires retraining for each new acquisition condition.

3.7.7 Assumptions and Limitations

A critical reading of TFPnP reveals several important assumptions and gaps:

1. **Gaussian denoiser mismatch.** The PnP denoiser $\mathcal{H}_\sigma$ is trained for Gaussian noise, but the *iteration noise* $\varepsilon^k = \bar{\boldsymbol{x}}^k - \boldsymbol{x}^*$ is not Gaussian in early iterations. The paper shows empirically that iteration noise becomes increasingly Gaussian as iterations proceed, and that PnP-ADMM achieves the best Gaussianisation ($R^2 = 0.972$)—partially justifying the choice, but not eliminating the mismatch.

2. **CT geometry limitation.** The TFPnP paper uses TorchRadon (**?**) with a simplified 30-view parallel-beam geometry. Generalisation to clinical cone-beam geometries is a gap that the present project, using LION/tomosipo (**??**), is positioned to address.

3. **Training cost.** Training the RL policy is substantially more expensive than training a supervised network, as it requires differentiating through many PnP iterations. Training times are not reported, making reproducibility assessment difficult.

4. **No convergence guarantee for the full system.** While convergence of PnP-ADMM under certain denoiser conditions is established (**??**), the addition of a learned adaptive policy breaks the assumptions of existing proofs. TFPnP is validated empirically, not theoretically.

5. **2D only.** All CT experiments are 2D (slice-by-slice), without addressing volumetric 3D reconstruction.

3.7.8 What the Paper Does Not Claim

TFPnP is presented as a *generic algorithmic framework*, not a CT-specific method. The paper does not claim optimality for any specific CT geometry beyond the 30-view parallel-beam benchmark, demonstrate clinical applicability on patient data, or address 3D reconstruction. These are the precise gaps that the present implementation project is directly positioned to investigate.

Chapter 4

Design and Implementation

4.1 Software Framework

4.1.1 Why LION?

LION (*Learned Iterative Optimisation Networks*) is an open-source Python toolbox developed by the Cambridge Computational Imaging (CIA) group at DAMTP, available at `https://github.com/CambridgeCIA/LION` (?). LION was chosen as the implementation framework for this project for several reasons:

- **Integrated CT pipeline.** LION provides end-to-end infrastructure for CT reconstruction research: dataset loading (including the Mayo Clinic Low Dose CT Grand Challenge dataset (?)), geometry configuration, sinogram simulation, classical reconstruction baselines (FBP/FDK, SIRT, TV), and learned reconstruction model training loops.

- **PyTorch native.** LION is built on PyTorch, allowing seamless integration of differentiable components—a prerequisite for the model-based RL training in TFPnP (?), which requires backpropagation through the PnP iterations.

- **Modular solver architecture.** LION's `Solver` abstraction separates the optimisation loop from the model definition, making it straightforward to implement a custom TFPnP training loop alongside the existing `SupervisedSolver` used by LPD (?).

- **TIGRE/tomosipo integration.** LION uses tomosipo and TIGRE under the hood, providing cone-beam CT operators not available in TorchRadon (which the original TFPnP paper used).

4.1.2 Why Tomosipo?

Tomosipo (?) is a Python library for defining CT acquisition geometries and computing forward/back-projection operators via GPU-accelerated ASTRA Toolbox backends. Its role in this project is:

- **Geometry abstraction.** A clean API for specifying parallel-beam and cone-beam scan geometries:

```
import tomosipo as ts
vg = ts.volume(shape=(1, 256, 256))
pg = ts.parallel(angles=30, shape=(1, 256))
A  = ts.operator(vg, pg)
```

- **Operator abstraction.** The resulting `A` object implements $\boldsymbol{x} \mapsto A\boldsymbol{x}$ (forward projection) and $\boldsymbol{y} \mapsto A^T\boldsymbol{y}$ (backprojection) as callable functions.
- **Differentiability.** Via `ts_algorithms`, the projectors can be wrapped in PyTorch's autograd framework.
- **Consistency with the paper's geometry.** The TFPnP paper uses a 30-view parallel-beam geometry with 182 detector pixels. Tomosipo can replicate this exactly.

4.1.3 Operator Abstraction

At the core of the CT pipeline is the system operator pair (A, A^T). In LION/tomosipo, these are implemented via the ASTRA Toolbox's CUDA-accelerated routines. The forward projection $\boldsymbol{y} = A\boldsymbol{x}$ simulates the physical measurement process described in **??**. The backprojection $A^T\boldsymbol{y}$ distributes each sinogram value back along its ray into the image volume.

4.1.4 Forward and Backprojection Implementation

For the $\boldsymbol{z}$-step of ADMM with quadratic data fidelity $D(\boldsymbol{y}, A\boldsymbol{x}) = \frac{1}{2}\|A\boldsymbol{x} - \boldsymbol{y}\|_2^2$, the proximal operator has the closed-form solution:

$$\operatorname{Prox}_{\frac{1}{\mu}D}(\boldsymbol{v}) = \left(\mu A^T A + I\right)^{-1}\left(\mu A^T\boldsymbol{y} + \boldsymbol{v}\right), \tag{4.1}$$

which requires an iterative solver. In the TFPnP implementation (?), a single gradient step is used as a tractable approximation:

$$\boldsymbol{z}^{k+1} \approx \boldsymbol{v} - \frac{1}{\mu}A^T(A\boldsymbol{v} - \boldsymbol{y}), \tag{4.2}$$

where $\boldsymbol{v} = \boldsymbol{x}^{k+1} + \boldsymbol{u}^k$. This is implemented as two tomosipo operator calls and is fully differentiable.

4.1.5 Tomosipo Exploration and Validation

Before implementing TFPnP, a systematic exploration of tomosipo was carried out (documented in `notebooks/01_tomosipo_fundamentals.ipynb`) to build operational understanding of the CT simulation pipeline. This step was essential for two reasons: first, the TFPnP paper uses TorchRadon (?) as its CT backend, whereas this project uses tomosipo/ASTRA; verifying that the same physical operations (forward projection, back-projection, noise simulation) behave equivalently in the new backend is a prerequisite for a valid reproduction. Second, every component of the TFPnP algorithm—the data-consistency $\boldsymbol{z}$-step (??), the reward function, and the training loop—depends on correct and efficient use of A and A^T, so fluency with the operator interface is necessary before writing any model code.

The exploration proceeded in three stages:

Stage 1: Geometry and operator fundamentals. Cone-beam and parallel-beam geometries were defined and compared. Cone-beam geometry (`ts.cone`) was used initially for general 3D exploration, as it represents the most common clinical CT configuration and exercises the full tomosipo API. The exploration then moved to 2D parallel-beam geometry (`ts.parallel` with a single detector row), which matches the TFPnP experimental setup: 30 projection angles over an $N \times N$ image with N detector pixels. The distinction matters because parallel-beam projections are shift-invariant (enabling analytical FBP inversion via the Central Slice Theorem), whereas cone-beam projections are not, requiring the FDK approximation (?). All subsequent project experiments use parallel-beam geometry to maintain consistency with the reference paper.

Stage 2: SIRT implementation from scratch. The SIRT algorithm was implemented manually using the tomosipo operator interface, following the update rule from ??. This served two purposes: (i) it verified that the forward operator A and its adjoint A^T behave correctly (the reconstruction converges to the ground truth for noiseless, fully sampled data); and (ii) it confirmed that the GPU-accelerated PyTorch path through tomosipo produces identical results to the NumPy path, which is important since TFPnP training requires GPU tensors throughout. The manual implementation was validated against `ts_algorithms.sirt` and ASTRA's internal `SIRT3D_CUDA`, with negligible differences in all cases.

Stage 3: Noise simulation and baseline reconstruction. Gaussian noise at the three levels used in the TFPnP paper ($\sigma \in \{5\%, 7.5\%, 10\%\}$) was added to sparse-view (30-angle) sinograms. Four reconstruction algorithms were then compared: FBP (via `tsa.fbp`), SIRT, Nesterov-accelerated least squares (NAG_LS), and TV minimisation (via `tsa.tv_min2d`). PSNR and SSIM were computed for each, establishing the baseline performance landscape against which TFPnP will be evaluated. The key observations were: (i) FBP from 30 views produces severe streak artefacts and low PSNR, confirming the ill-posedness described in **??**; (ii) TV minimisation substantially outperforms unregularised methods but introduces staircase artefacts; (iii) the quantitative gap between TV and the TFPnP results reported in **??** defines the improvement that a learned method must achieve. These baseline reconstructions also serve as the reference images for the qualitative comparison figures in **??**.

4.2 Dataset Pipeline

4.3 Model Architecture

4.4 Training Procedure

4.5 Baseline Implementations

Chapter 5

Evaluation

5.1 Experimental Setup

5.2 Baseline Comparison

5.3 Quantitative Metrics

5.4 Qualitative Analysis

5.5 Ablation Studies

5.6 Discussion

Chapter 6

Discussion

6.1 Summary

Chapter 7

Conclusion

7.1 Summary

7.2 Contributions

7.3 Limitations

7.4 Future Work

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Appendix A

Additional Technical Details